

The artifact selected for this category is a full-stack dashboard application, which was developed in CS 340 Advanced Programming Concepts. The application was designed for the fictional company, Grazioso Salvare, which was an international search and rescue training organization. The purpose of the dashboard was to allow users to filter and visualize data from animal shelters in order to identify potential candidates for their rescue training programs. The application was built through the use of Python, MongoDB, and the Dash framework. The system integrates a MongoDB database with a Python CRUD module to retrieve and manipulate data. The dashboard then displayed the data within an interactive table and provided a pie chart to visualize breed distribution as well as a map showing the location of selected animals. Users could then filter the data by rescue category, like water or mountain rescue, to allow for quick identification of suitable animals for the training programs. The artifact includes the dashboard implementation in Python as well as the CRUD module for communication with the MongoDB database.

This artifact was selected to be included within my ePortfolio as I feel it demonstrates various skills, including the ability to design and implement a full-stack and data-driven application that integrates a database with a dashboard created for users to improve readability of the data. This project presents a strong foundation in both backend development and data visualization. I believe that the artifact highlights a few software development skills, such as the prioritization of separating the application into distinct layers (CRUD Database, service, ranking, dash callbacks, etc.). Aside from that, the CRUD module showcases the ability to interact with a MongoDB database. The ranking algorithm also highlights the evaluation of data based on multiple criteria.

This artifact was significantly improved through several enhancements focusing on the database side of things. For instance, indexing was added to frequently queried fields like the breed or age of an animal. This was done to improve query performance/focus on database optimization.

Aggregation pipelines were added to compute breed distributions within MongoDB to shift some of the computational workload from the application layer, also in an effort to improve efficiency. Pagination was added to limit the number of records returned for each query to assist in scalability and response time when it comes to larger datasets. Validation and error handling were also strengthened in CRUD operations.

I believe that these enhancements successfully meet the course outcomes, such as the fact that the work demonstrates the ability to design and evaluate computing solutions through the use of appropriate tools and practices. It also shows the ability to implement efficient and scalable database operations. In addition, environment-based configuration and improved error handling show a security-focused mindset by protecting sensitive information and improving system reliability.

Throughout the process of enhancing this artifact, I gained a deeper understanding on utilizing database features in order to improve the overall performance and scalability of an application. I learned how to utilize aggregation pipelines to perform data analysis within the database and the benefit of indexing. One of the main challenges I encountered during this process was the integration of aggregation pipelines into the existing dashboard, as it required a rethinking on how the data was to be processed and displayed.